

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4619: Peripherals and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Write short notes on fixed and variable addresses. Explain the memory-mapped I/O with an appropriate scenario where port #02h is connected with a DIP switch and port #05h is connected with a colored LED. 5 + 5
(CO1)
(PO1)
- b) What is Harvard Architecture for microcontroller design? Explain the functions and applications of appropriate pins of ATmega16 microcontroller for analog I/O and external clock input operation. 2 + 6
(CO1)
(PO1)
- c) Differentiate between a microprocessor and microcontroller. What should be an extremely mega microcontrollers name if it is manufactured by ATMEL along with 512 Kbyte of flash memory? 5 + 2
(CO1)
(PO1)
2. a) What is the significance of input weights in an A/D or D/A convertor? Discuss the function of the comparator in the ADC. 5 + 5
(CO3)
(PO1)
- b) Consider a *Successive Approximation A/D* converter with $V_{in} = 0.7 \text{ Volt}$, $V_{ref} = 1 \text{ Volt}$, and 8-bit of resolution. Find the 8-bit digital output of this converter. Also, find the analog value for that digital output using the *Weighted Sum D/A* conversion method. 5 + 3
(CO4)
(PO1)
- c) Discuss the advantage(s) of R/2R ladder DAC over those that use binary weighted resistors? An 8-bit D/A converter produces $V_{OUT} = 0.25 \text{ Volts}$ for a digital input of 10000001. Find the value of V_{out} for an input of 11110000. 3 + 4
(CO2)
(PO1)
3. a) Suppose, an 8086 microprocessor is asked to address two 8255 ICs (i.e., 16th and 48th) sequentially and write control words at the control register of those ICs, respectively. In both 8255 ICs, Port-A is in Mode-2, Port-B is in Mode-1 as an output port and Port-C is handshaking mode. Now, to address the control registers of both the 8255 ICs, derive the sequential binary values of A7 – A0 pins and draw the respective control words. 5 + 5
(CO4)
(PO1)
- b) Write a structure of assembly language code for IC 8155 (i.e., compatible with IC 8085) to show a single bit input-output operation based on the I/O port scenario given in Question 3.a. 8
(CO2)
(PO2)
- c) Draw a timing diagram of a 8255 PPI when its Port-A is used as input port having interrupt request with a 8-bit data of all 0, Port-B is used as output port having no interrupt requested with a 8-bit data of all 1, and Port-C is in the handshaking mode. Also, derive the command register value. 5 + 2
(CO1)
(PO1)

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MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 50

CSE 4615: Wireless Networks

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. A microwave transmitter has an output of 0.1 W at 2 GHz. Assume that, the transmitter is used in a microwave communication system where the transmitting antenna is a parabola at a height of 81 m and the receiving antenna is a conical horn at ground level, each 1.2 m in diameter.
 - a) What is the gain of each antenna in decibels? Note that, the effective area, A_e of a parabolic antenna (with a face area A) is $0.56A$ and that of a conical antenna (with a mouth area A) is $\frac{2\pi A}{\lambda^2}$. 4
(CO2)
(PO1)
 - b) If the receiving antenna is located 20 km away from the transmitting antenna over a free space path, find the available signal power of the receiving antenna in dBm units. 4
(CO2)
(PO1)
 - c) What is the maximum coverage of this (Line-of-sight) propagation? Which height will you set for the receiving antenna to triple the maximum coverage? 2 + 2
(CO2)
(PO1)
 - d) In designing a communication system, the communication engineers need to estimate the effects of multipath fading and noise on the mobile channel. Describe the Rayleigh and Rician fading with the necessary diagrams. 4
(CO1)
(PO1)
2.
 - a) Explain the different modes of scanning in IEEE 802.11 WLAN. Discuss the advantages and disadvantages of these scanning modes. 4
(CO1)
(PO1)
 - b) One MAC Service Data Unit (MSDU) may be fragmented into multiple MAC Protocol Data Units (MPDU) to reduce the duration of channel occupancy in IEEE 802.11 WLAN. Discuss the benefits and drawbacks of the fragmentation process. 4
(CO1)
(PO1)
 - c) Consider an infrastructure-based 802.11 WLAN with two stations, station A and station B, both trying to transmit data packets to the Access Point (AP) within the same Basic Service Set (BSS) using the Distributed Coordination Function (DCF). The stations are within the range of each other and can hear each other's transmissions. In the first attempt, they choose the same random number as the backoff value from the contention window, $CW = 16$. However, in the second attempt they choose a different random number as the backoff value. Considering this scenario, list the sequence of events along with the necessary labeled diagram. 6
(CO1)
(PO1)
 - d) What is the disadvantage to the time domain backoff strategy? Describe Back2f (Backoff in frequency Domain) strategy and how it can overcome the disadvantage. 4
(CO1)
(PO1)
3.
 - a) What are the differences between multiplexing and multiple access techniques? Explain the Space Division Multiple Access (SDMA) technique. How can we enhance the channel utilization even after using the SDMA protocol? 4
(CO1)
(PO1)
 - b) Illustrate the effect of attenuation and noise on transmission power and comment on their relationship with the maximum transmission-range with a labeled bar chart. 4
(CO1)
(PO1)

- c) Describe the working procedure of the Timing Synchronization Function (TSF) in the Basic Service Set (BSS) and the Independent Basic Service Set (IBSS). Why TSF is required in IEEE 802.11 WLAN? 4
(CO1)
(PO1)
- d) Describe the architecture of Mesh network. Discuss the advantages and disadvantages of a Mesh network over a typical infrastructure network. 4
(CO1)
(PO1)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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DEPARTMENT OF BUSINESS AND TECHNOLOGY MANAGEMENT

Mid-Semester Examination
Course No. : HUM 4641
Course Title : Accounting

Summer Semester, A. Y. 2022-2023
Time : 1.5 hours
Full Marks : 75

Answer all 3 (three) questions. All questions carry equal marks. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1.

a) Explain accounting and users of accounting information with examples.

05 (CO1)
(PO1)

b) Describe monetary unit assumption and economic entity assumption with examples.

05 (CO1)
(PO1)

c) Trixie Maye started her own consulting firm, Matrix Consulting, on May 1, 2020. The following transactions occurred during the month of May.

15 (CO2)
(PO11)

May 1: Trixie invested \$7,000 cash in the business.

May 2: Paid \$900 for office rent for the month.

May 3: Purchased \$600 of supplies on account.

May 5: Paid \$125 to advertise in the County News Company.

May 7: Received \$4,000 cash for services performed.

May 8: Withdrew \$1,000 cash for personal use.

May 10: Performed \$5,400 of services on account.

May 12: Paid \$2,500 for employee salaries.

May 15: Paid for the supplies purchased on account on May 3.

May 16: Received a cash payment of \$4,000 for services performed on May 10.

May 20: Borrowed \$5,000 from the bank on a note payable.

May 22: Purchased equipment for \$4,200 on account.
- Instructions:

Prepare a tabular analysis of transactions using the following accounts:

Cash, accounts receivable, supplies, equipment, accounts payable, notes payable, owner's capital, owner's drawings, revenue and expense.
2.

Emily Valley is a licensed dentist. The business started on 1st April 2017. During the first month of the operation of her business, the following events and transactions occurred.

25 (CO2)
(PO11)

April 1 Invested \$20,000 cash in her business.

2 Paid office rent for the month \$1,100.

3 Purchased dental supplies on account from Dazzle Company \$4,000.

10 Performed dental services and billed customers \$5,100.

11 Received \$1,000 cash advance from Leah Mataruka for an implant.

20 Received \$2,100 cash for services performed from Michael Santos.

30 Paid secretary-receptionist for the month \$2,800.

30 Paid \$2,400 to Dazzle for accounts payable due.

Instructions:

- I. Journalize the transactions.
- II. Post journal accounts to the ledger accounts.

3. The trail balance for the Zarin Inc. on March 31, 2017 is as follows:

Account titles	Dr	Cr
Cash	4500	
Accounts Receivable	3200	
Supplies	2000	
Equipment	11000	
Accumulated Depreciation-Equipment		1250
Accounts Payable		2500
Unearned Service Revenue		550
Owner's Capital		12900
Owners Drawings	1100	
Service Revenue		6300
Salaries and Wage Expense	1300	
Miscellaneous Expense	400	
Total	<u>23500</u>	<u>23500</u>

Other data:

1. A physical count reveals only \$480 of roofing supplies on hand.
2. Depreciation for March is \$250.
3. Unearned revenue amounted to \$260 on March 31.
4. Accrued salaries are \$700.

Instructions:

- | | |
|---|--------------------|
| a) Complete the worksheet for Zarin Inc. | 13 (CO2)
(PO11) |
| b) Prepare an income statement for March, 2017 and balance sheet on March 31, 2017. | 08 (CO2)
(PO11) |
| c) Journalize the closing entries from the financial statements. | 04 (CO2)
(PO11) |

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FULL MARKS: 75

CSE 4643: Mobile Application Development

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) With proper justification, choose the most suitable approach for developing a web application from the following options: 8
(CO2)
(PO1)
- The classic use of HTML, CSS and JavaScript
 - Leveraging ReactJs

- b) Differentiate between var and let keywords using Code Snippet 1, and draw the dissimilarities with stack procedure. 8
(CO1)
(PO1)

```
1 var v1 = 10;
2 let l2 = 17;
3 const f1 = () => {
4     var v0 = 9;
5     let l0 = 8;
6     if(1){
7         let l1 = 7;
8         var v3 = 121;
9         console.log(l0);
10        console.log(l1);
11        console.log(v1);
12        console.log(l2);
13    }
14    //console.log(l1);
15    console.log(v3);
16 };
17 f1();
18 console.log(v0);
```

Code Snippet 1: A JavaScript code for Question 1.b

- c) What are the alternative methods of console.log() for the output method? 4
(CO1)
(PO1)

- d) Find the output of Code Snippet 2. 5
(CO2)
(PO1)

```
1 const names = ['atiq', 'shafin', 'rahman', 'edon', 'jafrin']
2
3 console.log(names.filter(nm => nm.length < 6))
```

Code Snippet 2: A JavaScript code for Question 1.d

2. a) IUTPS wants to launch a new mobile application to showcase photos. This application will have a few primary categories which are SHOOT-AT-IUT, Contests, and Winners. The categories are equivalent in hierarchy. All the categories should always be visible for quick access. IUTPS has requested a unique feature in which the title, SHOOT-AT-IUT, can be changed to something else, such as SHOOT-ON-16-December, based on the current contest. The contents are intended to be updated regularly. Since it is to showcase the photos, the application's UI is bound to be simple and efficient. Illustrate a mobile navigation pattern that is best for the given scenario with proper justification. 10
(CO4)
(PO2)
- b) Medication Tracker is a software that notifies users when they should take their medications. The developers used a side drawer navigation system positioned on the bottom. They added lots of primary categories with varying importance. The most important features were placed on the top, but sometimes scrolling is needed to access some of the features. Find the anti-patterns from the given scenario and discuss their respective solutions. 5
(CO3)
(PO2)
- c) State two reasons for which sidebars will never be considered as a persistent navigation pattern. 5
(CO1)
(PO1)
- d) Define and illustrate the Skeuomorphic pattern. 5
(CO1)
(PO1)
3. a) Find the output(s) of Code Snippet 3 for the following values of id variable: 5
(CO2)
(PO1)
- i. id = 8
- ii. id = 27

```

1 switch(true){
2     case (id%2)===0:
3         console.log('id is even');
4         break;
5     case (id%3)===0:
6         console.log('id is multiple of 3');
7     case (id%2) !== 0:
8         console.log('id is odd');
9         break;
10    default:
11        console.log('id is nothing of the above');
12 }

```

Code Snippet 3: A JavaScript code for Question 3.a

- b) Replace the placeholders with proper keywords of react native so that Code Snippet 4 will render the Figure 1. 5
(CO2)
(PO1)

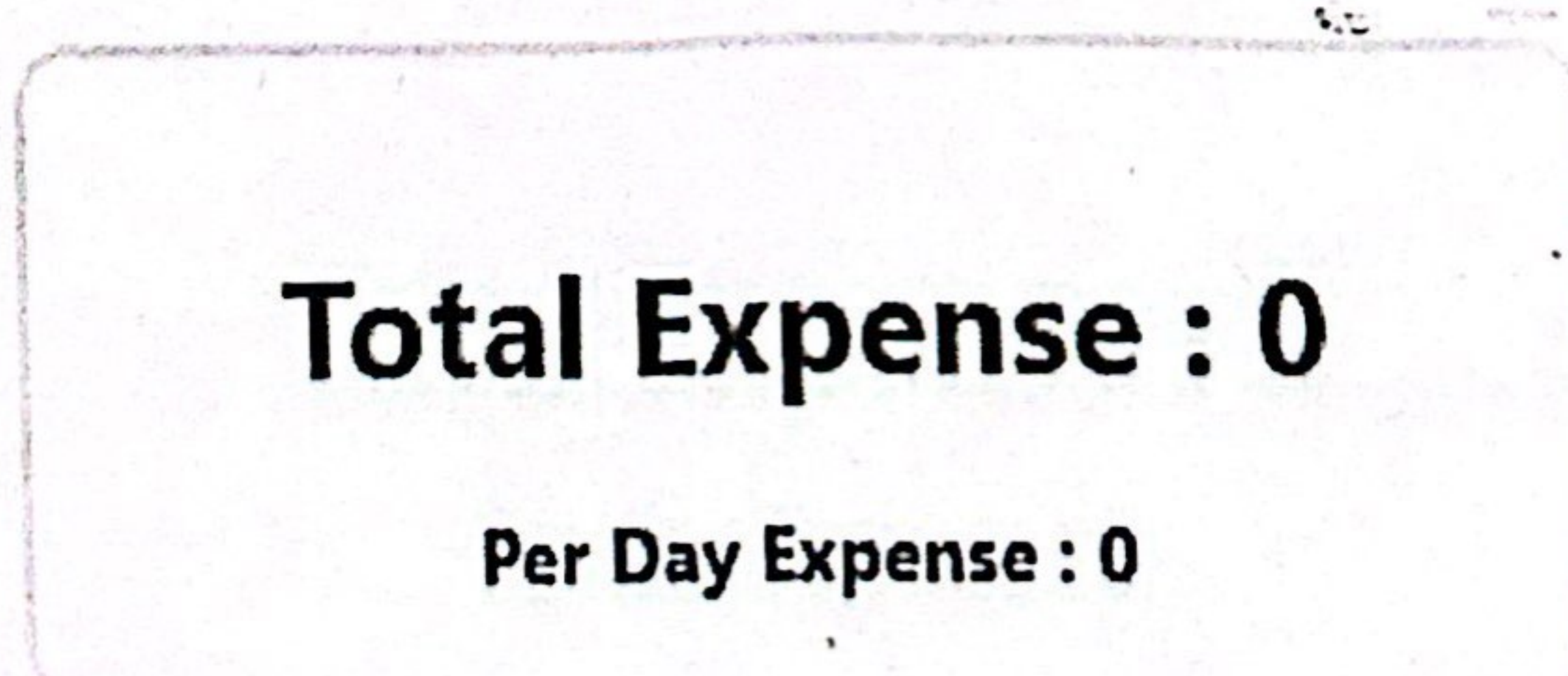


Figure 1: An image of a component for Question 3.b


```

1 import {Text,View,StyleSheet} from 'react-native'
2 import {useEffect,useState} from 'placeholder#1'
3 const CalculateExpense = ({expData}) => {
4
5   const [es,setEs] = useState(0)
6   const [pe,setPe] = useState(0)
7
8   'placeholder#2'(()=>{
9     if (expData && expData.length > 0){
10       let mn = new Date(expData[0].dt)
11       let mx = new Date(expData[0].dt)
12       let te = 0
13       expData.forEach((item,index)=>{
14         console.log(mn,mx,item.dt)
15         mn = Math.min(mn,new Date(item.dt))
16         mx = Math.max(mx,new Date(item.dt))
17
18         te += parseInt(item.exp)
19       })
20       setEs(te)
21       setPe((te / (( Math.abs(mx - mn) / (1000 * 60 * 60 * 24) )+1)).
22         toFixed(2))
23     } else {
24       setEs(0)
25       setPe(0)
26     }
27   },[expData])
28   return (
29     <View style={styles.timerContainer}>
30       <Text style={styles.elapsedTime}>'placeholder#3' : {es}</Text>
31       <Text style={styles.title}>Per Day Expense : {pe}</Text>
32     </'placeholder#4'>
33   )
34 }
35 export default 'placeholder#5'

```

Code Snippet 4: A React Native code for question 3.b

c) Discuss the following three terms with examples:

- i. Asynchronous Programming
- ii. Setup function of useEffect
- iii. Conditional Rendering of React Native

2 × 3
(CO1)
(PO1)

d) Mention the output of Code Snippet 5.

```

1 const numbers = [45,4,9,16,25];
2 let txt = '';
3 numbers.forEach(myFunction);
4 function myFunction(value){
5   txt += value;
6 }

```

4
(CO2)
(PO1)

Code Snippet 5: A JavaScript code for Question 3.d

e) Differentiate between states and props of React Native.

5
(CO1)
(PO1)

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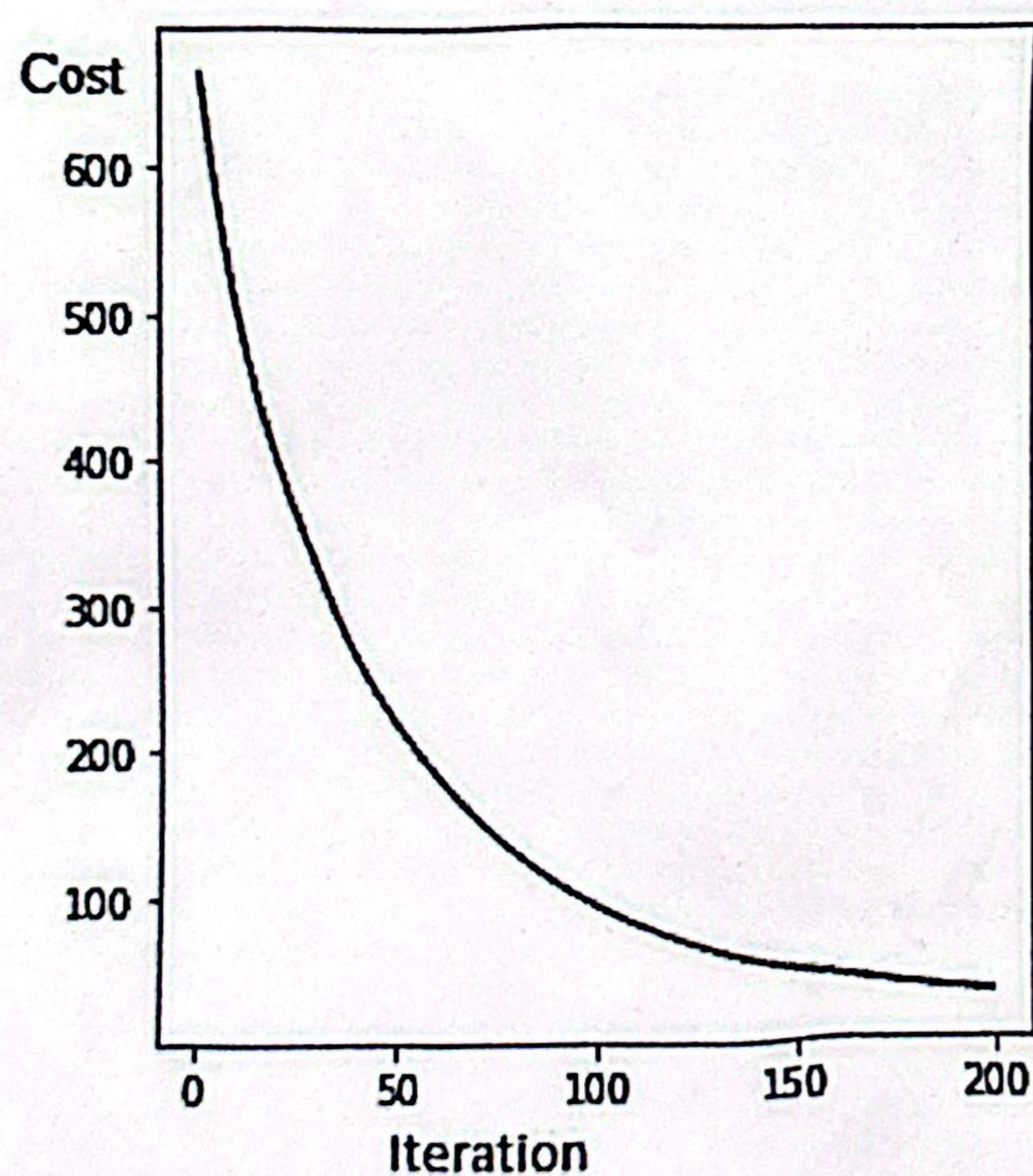
MID SEMESTER EXAMINATION
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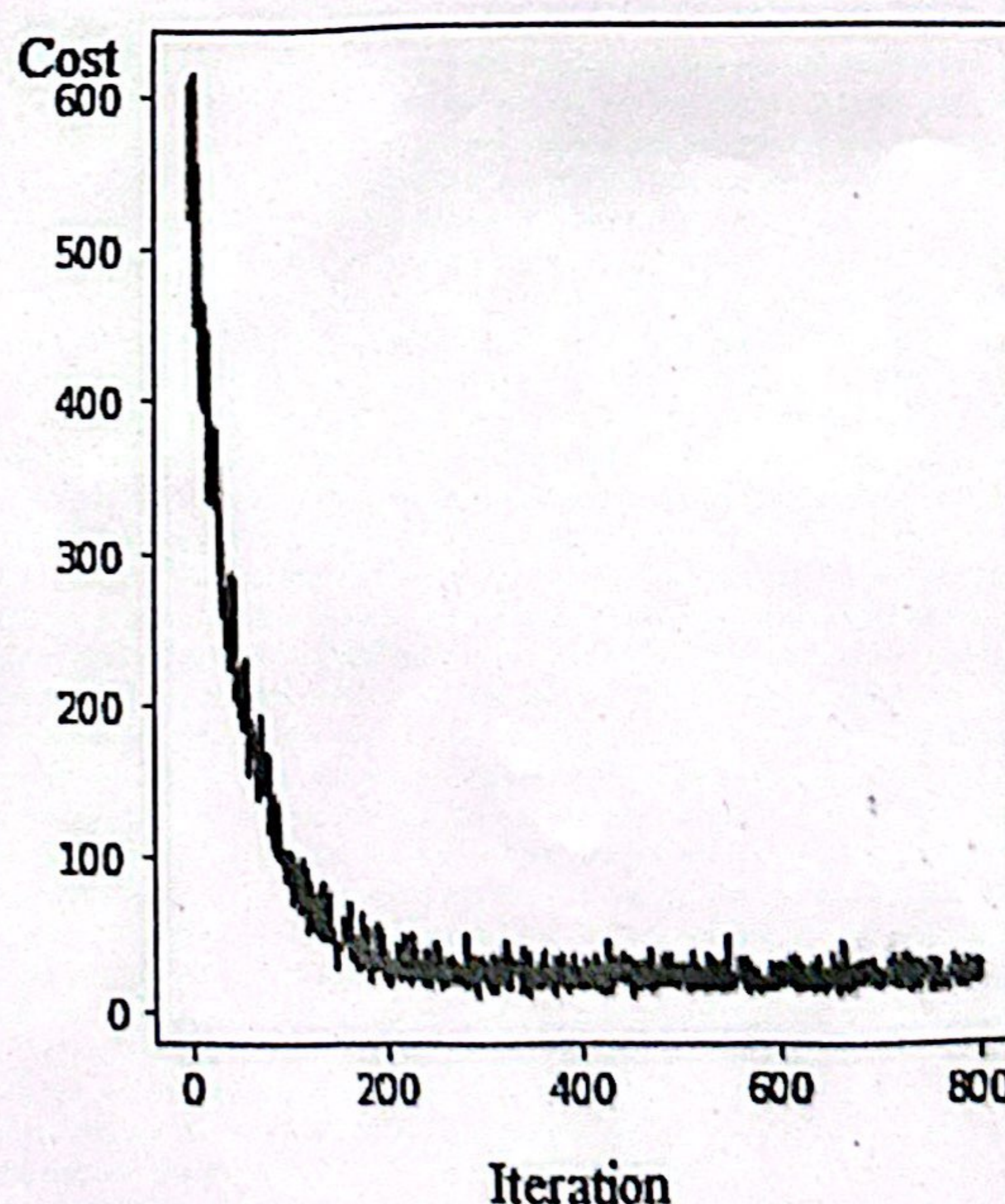
CSE 4621: Machine Learning

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Briefly explain the basic workflow of a typical supervised learning algorithm. 6
(CO1)
(PO1)
- b) With necessary figures identify why feature scaling is important before applying the gradient descent technique. Write the equation to rescale the range of a feature to a new scale $[-2, 3]$. 6 + 2
(CO1)
(PO1)
- c) Mathematically show that the binary cross-entropy function with L2 regularization term is convex. 5
(CO1)
(PO1)
- d) Consider the two different cost functions $J(\theta)$ plotted over many iterations as shown in Figure 1. Which graph corresponds to the use of Batch gradient descent (BGD) and which one corresponds to use of Mini-Batch gradient descent (mBGD). Justify your choice. 6
(CO2)
(PO2)



(a) Cost values for case 1.



(b) Cost values for case 2.

Figure 1: Cost function curves for Question 1.d

2. a) For a binary classification model, we predict for the positive class (i.e., $y = 1$) when $P(y = 1|x) \geq 0.5$, where a logistic function is used to convert to a probabilistic value. Mathematically show that this model is linear. 7
(CO1)
(PO1)
- b) How can you extend a binary classifier into a multi-class classifier considering a one-versus-rest strategy? What are the limitations of this design? Draw figures as necessary. 6 + 2
(CO1)
(PO1)

- c) Consider a rare scenario where the d -dimensional feature vectors contains binary features, i.e. $x \in \{0, 1\}^d$, where x_1 is mostly 0, and happens to be 1 in the training set with only positive class ($y = 1$) samples. Roughly estimate the value of $\hat{\theta}_1$. Justify your answer 5
(CO1)
(PO1)
- d) Is it possible to get a closed form solution for the parameter $\hat{\theta}$ when the cost function $J(\theta)$ is the log-loss function for binary classification? Explain why. 5
(CO1)
(PO1)
3. a) Your teammates are training a neural network for a classification task using a feed-forward architecture with one hidden layer. They decide to tune the number of neurons in the hidden layer using cross-validation, and compare train, validation and test error. They run cross-validation for numbers of neurons in the hidden layer of [50, 100, 200]. Upon examining the test error curve, you notice that the test error increases when the number of neurons decreases beyond 50. After evaluating the models, they choose 200 neurons in the hidden layer because it resulted in the lowest test error. 5
(CO1)
(PO1)
- This observation suggests that there may be an optimal number of neurons that was not explored. You suggest extending the search to include additional values for the number of neurons, such as [250, 300, 350], to ensure a more comprehensive search for the best-performing neural network architecture.
- Do you believe you gave the right recommendation to your team considering the behavior of the test error? Justify your answer.

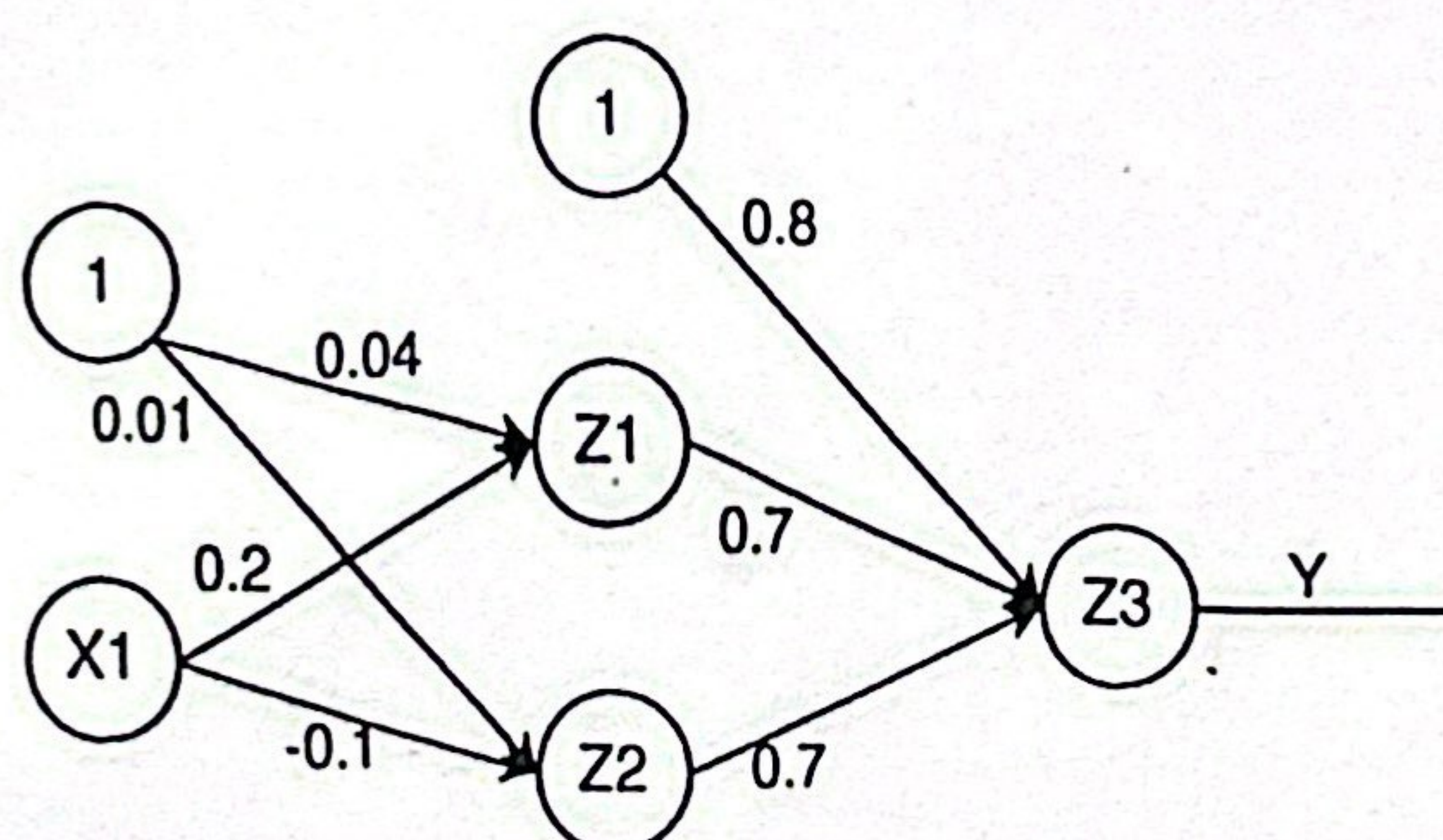


Figure 2: Neural Network for Question 3.b

- b) Consider the 2-layer neural network with its weights as shown in Figure 2. In the hidden and output layers, sigmoid activation functions are used. For a sample input $x = 0.7$, the actual output is $y = 1$. Answer the followings: 2 +
6 + 2
(CO1)
(PO1)
- Compute the predicted output $\hat{y}$.
 - Find the updated weight matrix $W^{[1]}$ after one epoch.
 - Also, find the updated bias $b^{[1]}$.
- c) Consider a very simple neural network with N layers that takes in a scalar input $x = a_0$, and for each layer $i \in \{1, 2, \dots, N\}$, we have 3 +
2 + 5
(CO1)
(PO1)
- $$z_i = w_i * a_{i-1} + b_i \quad (1)$$
- $$a_i = \sigma(z_i) \quad (2)$$

Note that w_i, b_i, z_i, a_i are all scalars. Answer the followings:

- Using the chain rule of calculus, give the mathematical expression for $\frac{\partial a_i}{\partial a_{i-1}}$. Your expression should only contain $\sigma'(z_i)$ and w_i .
- Since the derivative of the sigmoid function $\sigma'(\cdot)$ is at most $\frac{1}{4}$. Give an upper bound value for $\frac{\partial a_N}{\partial w_1}$.
- From Question 3.c)ii, how can you prove that for a neural network with too many layers, this deep network will lead to slow or unstable learning?

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SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4631: Digital Signal Processing

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Illustrate the block diagram of the discrete-time system given as follows in Equation 1: 11
$$y(n) = x_1(n) + 2x_2(n - 1) + 4x_1(n + 1)x_2(-n - 1) + x_3(n)y(n - 1) \tag{1}$$
 (CO1)
(PO1)
b) Suppose an input signal, $x(n)$, is passed through a system with impulse response, $h(n)$, producing the output, $y(n)$. The input signal and impulse response is given as follows: 7
$$x(n) = \{1, -2, 2, -3, 3, -3\}$$

$$h(n) = \{-1, 2, -3, 4\}$$

Here, both $x(n)$ and $h(n)$ start from the origin. Compute the output $y(n)$. (CO2)
(PO1)
c) Refer to the input signal given in Question 1.b. For the same input, find the impulse response of the system that produces the following output: 7
$$y(n) = \{-4, 11, -16, 23, -27, 29, -18, 9, -3\}$$
 (CO2)
(PO1)
2. a) Explain why we prefer digital signal processing instead of directly performing analog signal processing using various devices like transistors, op-amps etc. 8
(CO1)
(PO1)
b) Consider the following analog sinusoidal signal:
$$x_a(t) = 2 \sin(200\pi t)$$

i. The signal is sampled with a sampling rate $F_s = 300$ samples/s. Determine the frequency and the fundamental period of the resulting discrete-time signal $x(n)$. 6
(CO2)
(PO1)
ii. Find the minimum sampling rate F_s for which the signal can reach its peak value of 2. 5
(CO2)
(PO1)
c) A discrete-time signal given as $x(n) = 12.75 \cos(\frac{\pi n}{10})$ is quantized with a resolution $\Delta = 0.1$. 6
Find the minimum number of bits required for coding in the A/D converter. (CO2)
(PO1)
3. a) Suppose $x(n)$ is a signal with 64 points in the time domain.
i. Provide the justifications regarding the number of components and the number of points in each component in the frequency domain representation of $x(n)$. 6
(CO3)
(PO2)
ii. Describe the different ways of representing the horizontal axis of the frequency domain, and also how to convert from one representation to another. 4
(CO3)
(PO2)
iii. Explain why we usually select the number of samples N in the time domain signal to be a power of 2. 4
(CO3)
(PO2)

- b) The following input-output pairs have been observed during the operation of a time-invariant system:

6
(CO2)
(PO1)

$$x_1(n) = \{1, 0, 2\} \xrightarrow{\tau} y_1(n) = \{0, 1, 2\}$$

$$x_2(n) = \{0, 0, 3\} \xrightarrow{\tau} y_2(n) = \{0, 1, 0, 2\}$$

$$x_3(n) = \{0, 0, 0, 1\} \xrightarrow{\tau} y_3(n) = \{1, 2, 1\}$$

Here, the leftmost point is the origin for each of the signals and τ represents the system. Provide justifications regarding the linearity of the system.

- c) Examine the following systems and determine whether they are static or dynamic, linear or nonlinear, time invariant or time variant, causal or noncausal, and, stable or unstable.

5
(CO1)
(PO1)

- i. $y(n) = \sum_{k=-\infty}^{n+1} x(k)$
- ii. $y(n) = x(2n)$
- iii. $y(n) = |x(n)|$
- iv. $y(n) = x(n) \cos(\omega n)$
- v. $y(n) = x(n) + nx(n+1)$